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**80** Pages
27.6 cm x 21.2 cm

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# EXERCISE BOOK
# CAHIER D'EXERCICES

**NAME/NOM** Cole Shanks

**SUBJECT/SUJET** ELEC 315

ASSEMBLED IN CANADA WITH IMPORTED MATERIALS
ASSEMBLÉ AU CANADA AVEC DES MATIÈRES IMPORTÉES

12107

# Lecture 0

9/03/19
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Course Overview:

1 midterm → 25%
1 Final → 50%
Assignments throughout → 15%
Participation (Iclicker) → 10%

Japan Solar Workshop → 10-3:00pm Kaiser

Office hours:

Tuesday → 11:00 - 12:00
4:00 - 5:00

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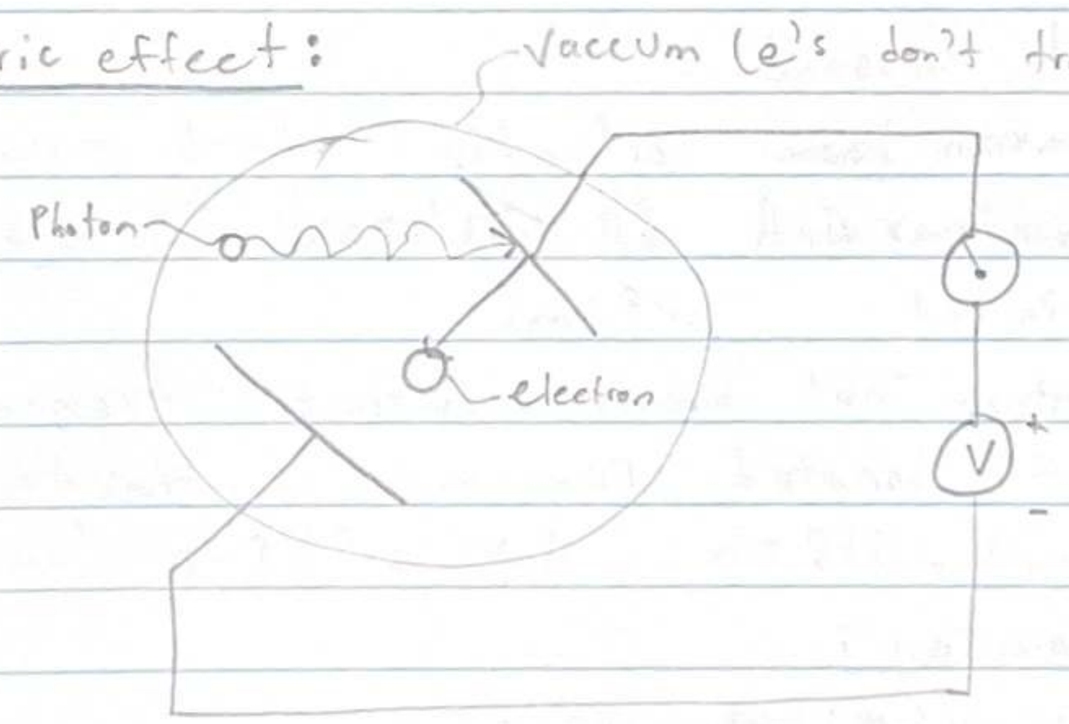
# Lecture 1

Photoelectric effect:




Conservation of energy: energy of incoming photon = energy for electron to get out + Kinetic energy for the electron

$KE_{Max} = -e \cdot \theta + h\nu$

h → Plank's constant
ν → Frequency

Photons:

Energy $= h\nu$

(Momentum) $p = \frac{h}{\lambda}$

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# Lecture 2

Conservation of energy: (assuming perfect collision

Photon (energy) = work function + Kinetic energy
- we use voltage to slow down approaching electrons

∴ Photon (energy) = work function + applied voltage
(at the point where current cuts off)

Heisenberg Uncertainty Principle:

① Perfect sinusoid
- momentum known perfectly
- $(-\infty, \infty)$ can't be localized to a point/region

② Wave Packet
- momentum not known (multiple frequencies)
- $(-x, x)$ located somewhere in the region X

Basic Equations:

① $E = h\nu$ ($\nu$) Frequency
($h$) Planck's constant

② $\lambda = \frac{h}{p}$

$\sin(kx - wt)$ | $k = \frac{2\pi}{\lambda}$ | $w = 2\pi f$ | $v = \lambda f$

Kinetic energy:

$KE = \frac{p^2}{2m}$ | $KE = \frac{h^2}{2m\lambda^2}$ | $KE = \frac{-\hbar^2}{2m\Psi} \cdot \frac{d^2\Psi}{dx^2}$

Shrödinger Equation:

$$j \cdot \hbar \cdot \frac{d\Psi}{dt} = \frac{-\hbar^2}{2m} \cdot \frac{d^2\Psi}{dx^2} + U \cdot \Psi$$

$$E(n) = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

6/19

# Lecture 3

$$\hbar = \frac{h}{2\pi}$$

(More confined → more discrete)

$\boxed{\text{1D Confinement}}$: $E(n) = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$

(less discrete energy levels)

$\boxed{\text{2D Confinement}}$: $E(n_x, n_y) = \frac{h^2}{8m}\left(\frac{n_x^2}{x_0^2} + \frac{n_y^2}{y_0^2}\right)$

Atomic Orbitals:

$n = 1$ (s orbital) $n = 2$ (P orbitals) $n = 3$ (D orbitals)

(Degrees of degeneracy) = (# of wave functions @ specific energy level)

# Lecture 4

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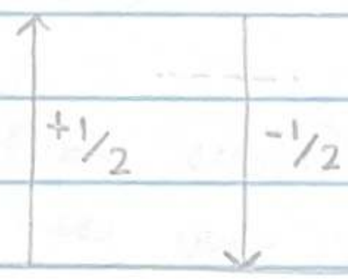
Covalent Bonding: (Bringing two H atoms close together)

- Energy levels split into new more levels
- Solid structure of many atoms can start to have a continuous looking number of energy levels (why solar cell can absorb anywhere)

Electron Spin:




- Quantized (up or down)

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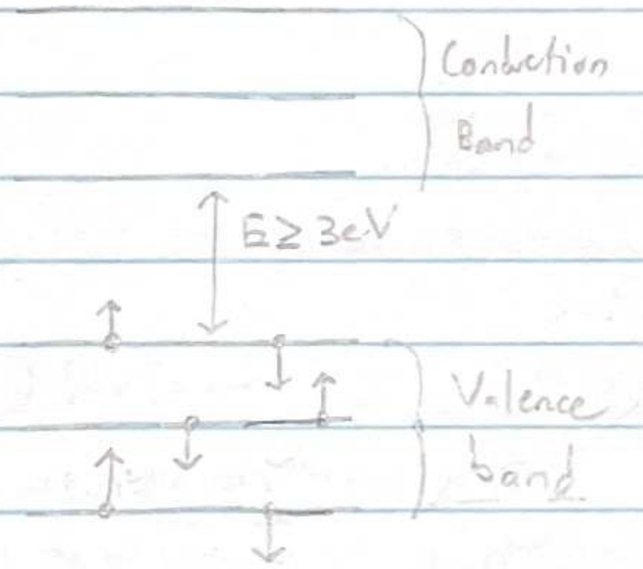
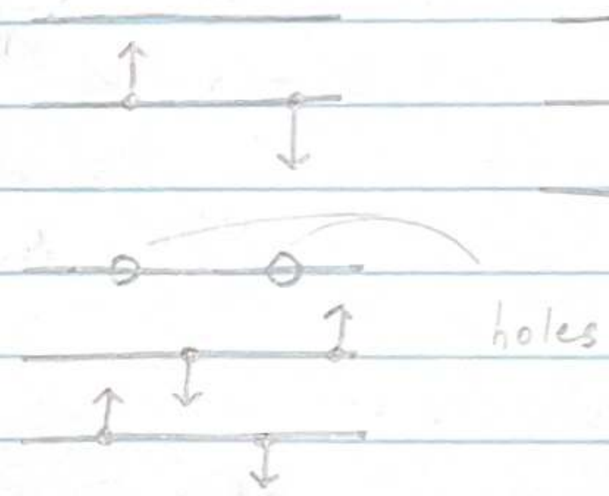
# Lecture 5 (solids and Semiconductors)

SLIDE SET

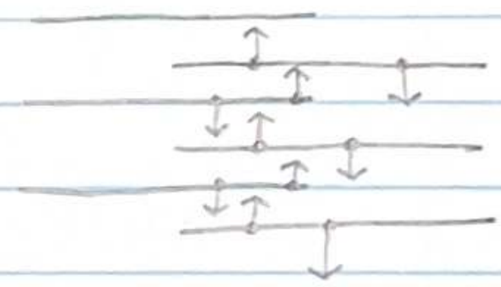
Insulator:




Semi Conductor:




Conductor (metal):



Q6 c-d (6-c-d) total energy = kinetic energy

Q8g Assume $n_x = n_y = n_z = 1$ (ground state) assume material is silicon

Polycrystalline:

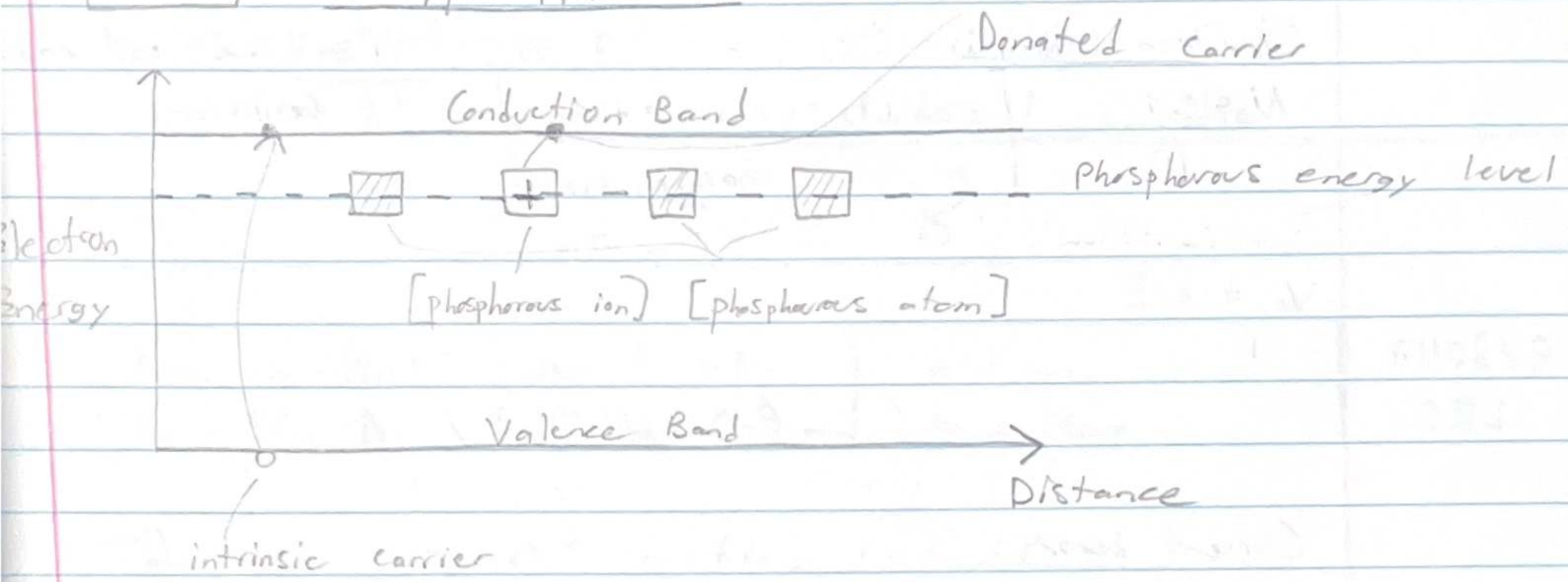
Grain boundaries non-conductive

SS11 'i' subscript means intrinsic ie: undoped

$n_i = p_i$ → # holes in valence band
→ # e's in conduction band

# Lecture 6

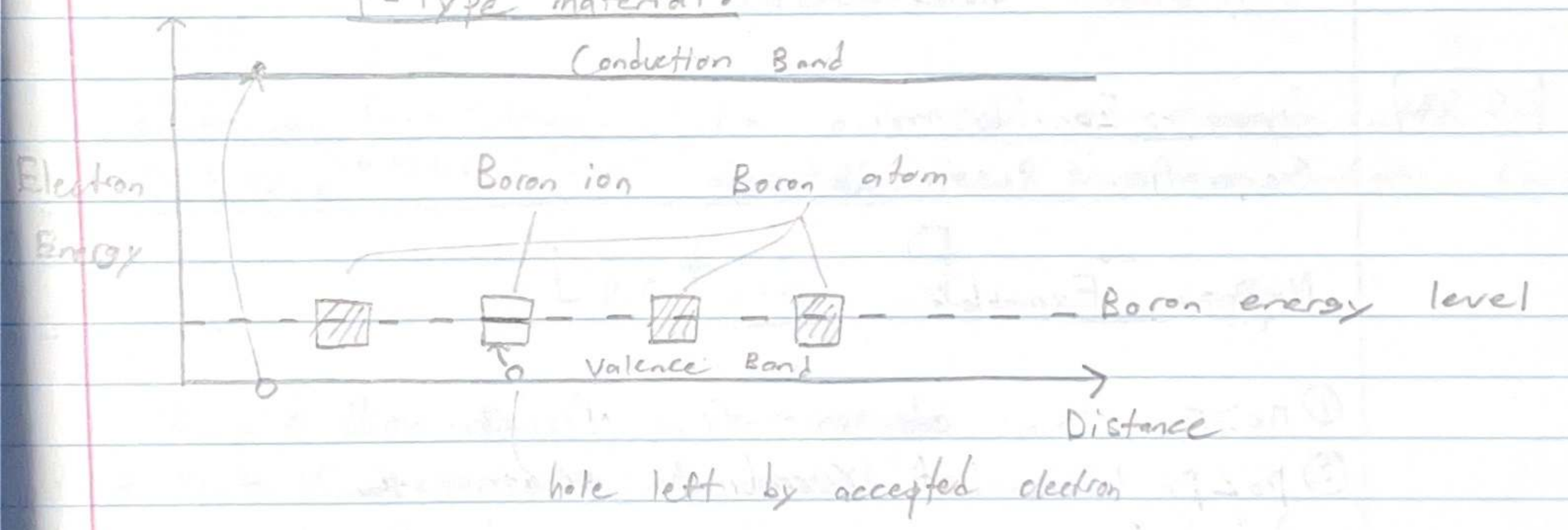
SS 15 **n-Type Material:**




**Electron effective mass:**

$M_0 = 9.1 \times 10^{-31}$ kg [Free electron mass]

**P-Type material:**




**Current:**

$J = \sigma E$ (Current density = Conductivity · Electric field)

**What effects Conductivity:**

1. time between collisions
2. electron mobility (effective mass)
3. # of carriers per unit volume

$$\text{Current} = \frac{L \cdot A \cdot n_0 \cdot q_e}{T} \quad \text{where: } \frac{L}{T} = V_e$$

Electron Mobility:

$$N = \frac{eT}{m} \qquad \boxed{\text{mobility} = \frac{e_{charge} \cdot \text{time}}{\text{mass}_{effective}}}$$

T = time between collisions

$$V_e = NE$$

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# Lecture 7

Current density:

$$J = e\mathcal{E}(n_0 N_e + p_0 N_h)$$

where: $n_0$ = electron density in conduction band
$p_0$ = hole density in valence band
$N_e$ = electron mobility
$N_h$ = hole mobility

SS 52

Dynamic Equilibrium:

Generation = Recombination

N-Doping Example:

① $n_0 > n_i$ (Donor electrons from Phosphorus)
② $p_0 < p_i$ (Rate of recombination greater than at intrinsic)

SS 60

Rate of Generation: [@ Equilibrium] $R = G$

$$G = f_g(\text{Temp})$$

Rate of Recombination:

$$R = f_r(\text{Temp}) n_0 p_0$$

$$n_0 p_0 = n_i p_i$$

$$\therefore \boxed{n_0 p_0 = n_i^2}$$

Density of States: (3D) [how many states per unit volume at the given energy level (E)]

$$g_c(E) = \frac{(m_e^*)^{3/2}}{\pi^2 \hbar^3} \sqrt{2(E - E_c)}$$

conduction band edge

# Lecture 8

$n_0 = N_D - N_A$ (when $N_D > N_A$)
$p_0 = N_A - N_D$ (when $N_A > N_D$)
} @ Room temperature

Then, use $n_0 p_0 = n_i^2$ to find $n_0, p_0$

Counting electrons:

$$n = \int_{CB} g_c(E) \cdot f(E)\, dE$$

where:
$n$ = total # e's in conduction band
$g_c(E)$ = Density of states
$f(E)$ = occupancy probability

Boltzmann Distribution: (can only use this approximation above the fermi level ie: conduction band)

$$f(e) \approx e^{-(E - E_F)/kT}$$

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# Lecture 9

$N_D$ = # donor atoms per unit volume
$N_A$ = # acceptor atoms per unit volume

n-doped material:
$n_0 = N_D$ (@ room temperature)

p-doped material:
$p_0 = N_A$ (@ room temperature)

## Couting holes:

$$P_o = \int_{\text{Bottom VB}}^{\text{Top VB}} g_v(E)(1-f(E))\,dE$$

## 3D box:

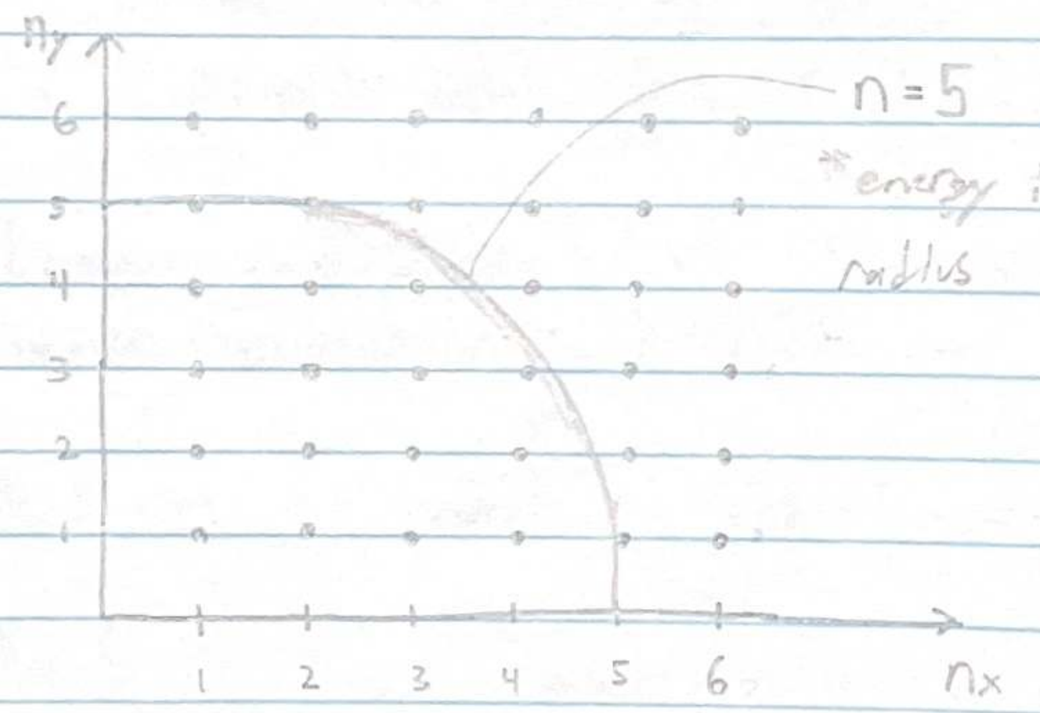
$$E = \frac{h^2}{8m}\left[\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}\right]$$

## 2D box:

if ($L_z \ll L_x$ and $L_y$) and ($L_x = L_y$)

$$E_{2D} = \frac{h^2}{8mL_x^2} \cdot [n_x^2 + n_y^2]$$

## Find density of States:




$n = 5$ $[n^2 = n_x^2 + n_y^2]$

*energy is constant along this radius

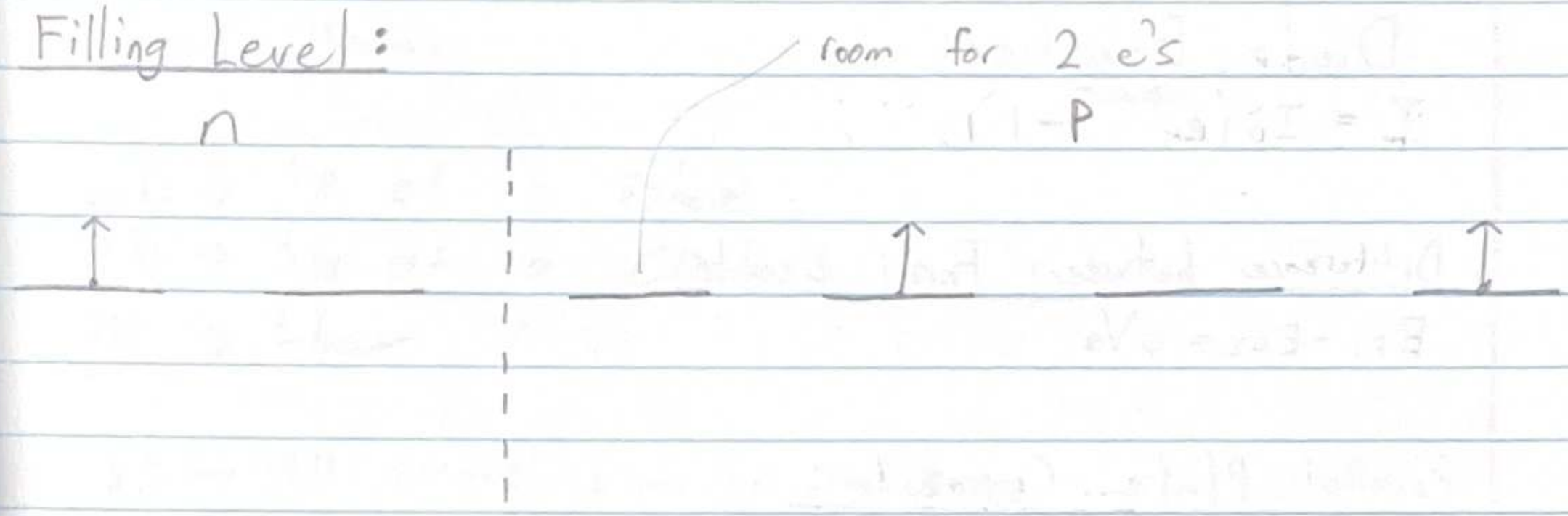
$$N = \frac{\pi n^2}{2}$$

## Intrinsic e's:

$n_i^2 = 10^{20}$

# Lecture 10

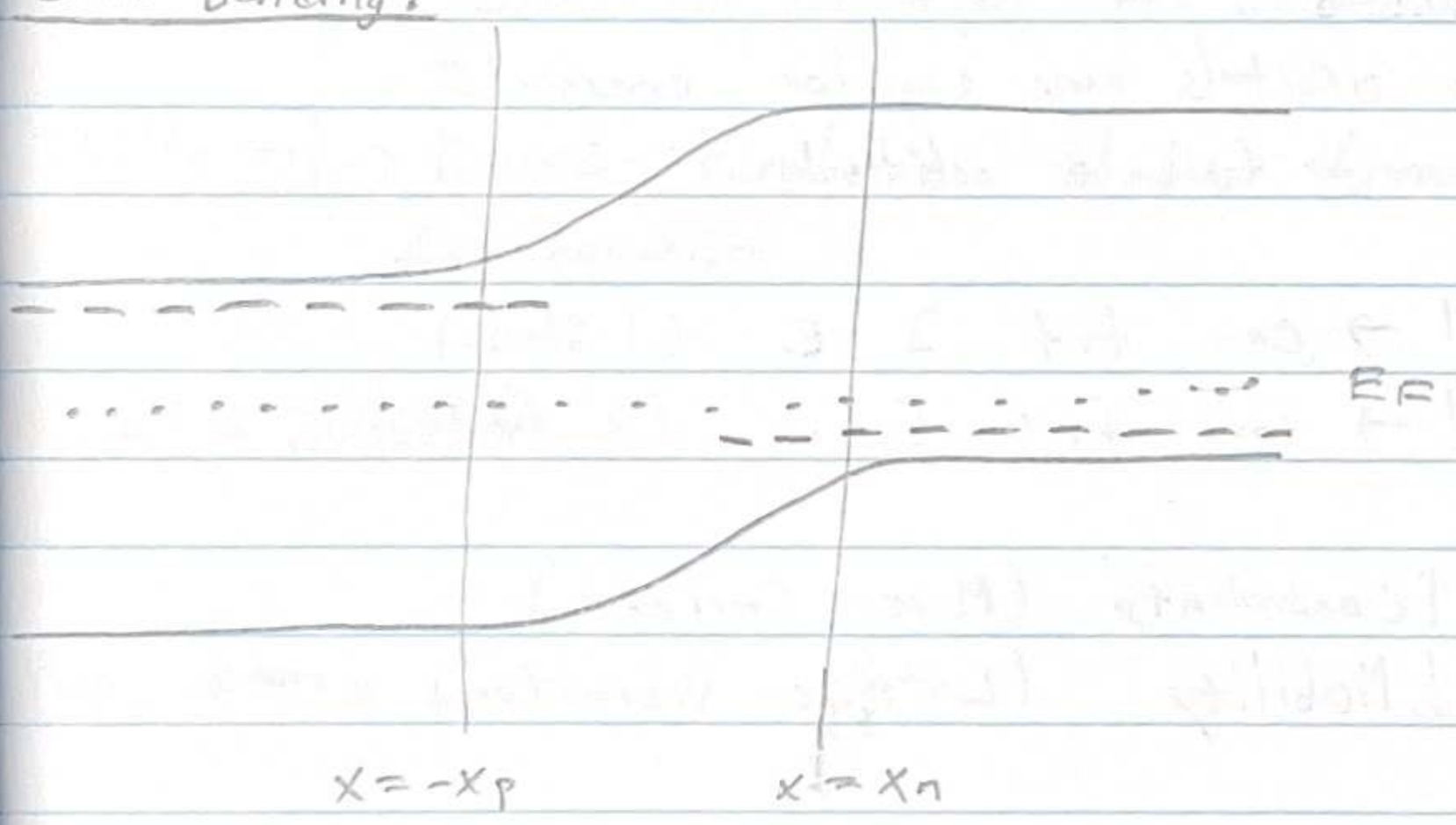
Filling Level:




$E_F = \frac{1}{4}$ $\qquad$ $E_F = \frac{2}{8} = \frac{1}{4}$

# Lecture 11

Band Bending:




SS 173 $\quad U(x) \propto \frac{+e\rho}{2\epsilon} x^2$

SS 179 $\quad$ ASN3 Q3 Derivation

# Lecture 12

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Diode Equation:

$I = I_0 (e^{eVa/RT} - 1)$

Difference between Fermi Levels:

$E_{F1} - E_{F2} = eVa$

Parallel Plate Capacitor:

$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$

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# Lecture 13

Review orbitals:

2s and 2p orbitals very similar energies

ie: Berrillyium → full p orbital

S orbital → can fit 2 e (1 state)

P orbital → can fit 6 e (3 states)

Intrinsic Material

↑Temp → ↑Conductivity (More carriers)

↑Temp → ↓Mobility (Lattice vibrations = more collisions)

*But $n_i >> N_e$

Doped Material

↑Temp → ↓Conductivity

↑Temp → ↓Mobility

$n_0 >> n_i$ so $n_0$ = Fixed ∴ $N_e$ dominates

↑Temp → ↑Conductivity (Intrinsic Semi-conductor)

↑Temp → ↓Conductivity (Metal)

↑Temp → ↓ → ↑ Conductivity (Doped semiconductor)

$\boxed{\text{Midterm}}$ → Derive either 1D, 2D or 3D $g(E)$ on midterm

3D → 1/8 of a sphere
2D → 1/4 of a circle
1D → Line

3D → All terms similar in size
2D → One term much bigger (ie: small L)
1D → Two terms much bigger (ie small L's)

Silicon Band Gap = 1.12eV

$\boxed{\text{Midterm}}$ → One PN Junction Band diagram (either equ, Fwd, reverse)

$\boxed{\text{Midterm}}$ → Derive Parabolic shape from depletion Approximation

$KT = 0.025V$

# Lecture 14

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** Common Exam Question → SS. 24

11/6/19
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# Lecture 15

Polarization:

$P = P_0 e^{-t/\tau} \longrightarrow P(\omega) = \dfrac{P_0}{1/\tau + j\omega}$

Susceptibility:

$$X(\omega) = \frac{P(\omega)}{\epsilon_0 E_0} = \frac{X_0}{1/\tau + j\omega}$$

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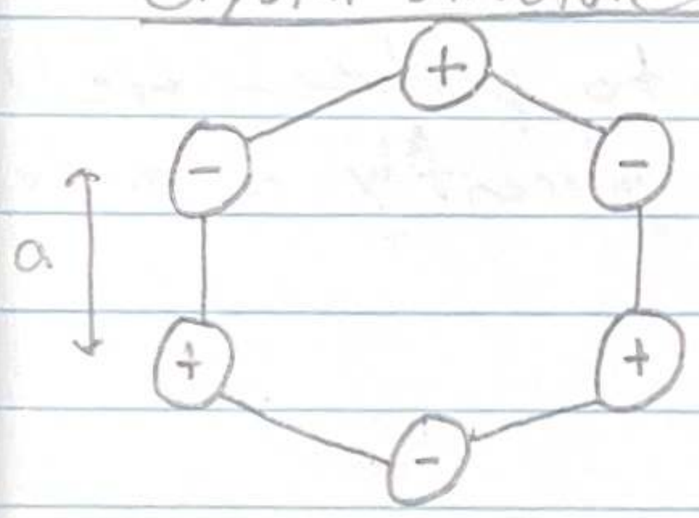
# Lecture 15

$\text{Energy} = \frac{1}{2} C V^2$

$eE\lambda \geq E_g$ (For electron to be promoted)

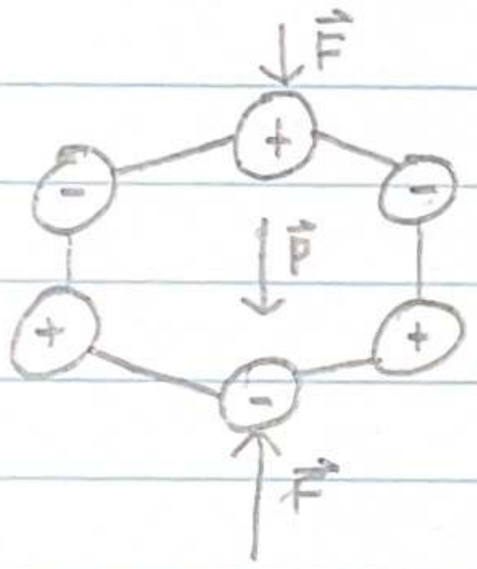
# Lecture 16

## Crystal Structure A:




- No dipole moment
- $P = 0$

## Crystal Structure (B) [Compressed]:




- Net dipole moment
- $\vec{P}$ is non-zero

## Direct Piezoelectric Equation:

$P_x = d_x T_x$ where:

$P_x$ = Polarization

$T$ = applied stress $\vec{F}/\text{Area}$

$d_x$ = piezoelectric constant

$$\frac{\Delta L}{L} = d E$$

$$E = \frac{\Delta L}{dL}$$

$$\frac{E}{L} = \frac{\Delta L}{dL^2}$$

$$V = \frac{\Delta L}{dL}$$